SCIENCE EXPLORER / GRADE 8

Small particles. Big changes.

Physical and chemical changes

An original lesson, revision guide and 100-question practice book

INDEPENDENT EDUCATIONAL PROTOTYPE

Informed by the KICD Grade 8 Integrated Science curriculum design, revised 2024, strand 1.0, sub-strand 1.2. The curriculum allocates 22 lessons to this sub-strand. This compact resource is a starting point, not a complete replacement for those lessons.

This is newly written teaching and practice material, not a reproduction of the curriculum, not an official KICD publication and not endorsed by KICD. Teacher review is required before classroom use.

Source location: uploaded PDF page 7 of 14, showing document pages 15-16 of 37 (printed curriculum pages 3-4). The associated assessment rubric is visible on uploaded PDF page 8.

Official curriculum listing: https://kicd.ac.ke/cbc-materials/curriculum-designs/grade-eight-designs/#category10

How to use: in the HTML, start with a short animation, make a prediction, and explore the optional explanation. Play, pause, step forward or scrub at your own pace. All five animations work offline. Use this PDF for deeper reading and practice; check the answer key after attempting questions.

Safety: no flames, hot liquids, unknown chemicals, tasting or direct smelling. Heating experiments require a teacher and suitable laboratory controls.

Your learning route

1. Particles & states: Describe how particles behave in solids, liquids and gases.

2. Changes of state: Explain state changes and interpret a simple heating curve.

3. Purity & temperature: Explain typical effects of impurities on melting and boiling.

4. Physical or chemical?: Classify changes using evidence of new substances.

5. Science in daily life: Apply state-change ideas safely and explain useful changes.

Coverage note: lessons 1-4 address particle behaviour, heating curves, impurities and classification; lesson 5 develops applications and appreciation through everyday examples. Practical skills and classroom collaboration still require teacher-led assessment.

1. Particles & states

Describe how particles behave in solids, liquids and gases.

Matter is made of tiny particles. A solid has closely packed particles that vibrate around fixed positions. This gives it a fixed shape and approximately fixed volume.

In a liquid, particles remain close together but can move past one another. A liquid flows, keeps approximately the same volume and takes the shape of its container.

Gas particles are widely separated and move randomly. A gas spreads to fill its container and is much easier to compress than a liquid or solid.

Diffusion is the net spreading of particles from a region of higher concentration to a region of lower concentration because of random motion. It can happen without stirring.

Heating generally increases particles' average kinetic energy when temperature rises. Particles themselves do not get bigger when a substance expands.

Think it through

Explain why air can be squeezed in a sealed syringe more easily than water. Use particle spacing, not particle size.

2. Changes of state

Explain state changes and interpret a simple heating curve.

Melting changes a solid to a liquid; freezing changes a liquid to a solid. Evaporation and boiling change liquid to gas. Condensation changes gas to liquid. These are physical changes: the substance keeps its chemical identity.

Evaporation happens at a liquid's surface and can occur below its boiling point. Boiling occurs throughout the liquid when conditions allow vapour bubbles to form and grow.

For pure water at standard atmospheric pressure, melting occurs at about 0 degrees Celsius and boiling at about 100 degrees Celsius. Pressure matters: water can boil at a lower temperature at higher altitude.

On an ideal heating curve for a pure substance at constant pressure, sloping sections show temperature increasing within one state. Flat sections show melting or boiling while energy changes particle arrangement and separation rather than increasing average kinetic energy.

A heating curve's horizontal axis can represent time only when the heating rate is suitably controlled. Our diagram uses energy supplied and is a simplified model, not measured experimental data.

Think it through

Ice is melting while energy is still supplied. Why can its temperature stay steady until the ice has melted?

Sketch a heating curve: label temperature on the vertical axis and energy supplied on the horizontal axis. Draw rising solid, liquid and gas sections, with a melting plateau and a boiling plateau between them. For pure water at standard atmospheric pressure, label the plateaus approximately 0 and 100 degrees Celsius. This is an idealisation.

3. Purity & temperature

Explain typical effects of impurities on melting and boiling.

A pure substance contains a single chemical substance. Under fixed conditions it has characteristic melting and boiling behaviour. A mixture contains more than one substance.

A dissolved non-volatile solute such as salt generally raises water's boiling point compared with pure water at the same pressure. 'Non-volatile' means the solute does not readily enter the vapour under these conditions.

Suitable dissolved impurities generally lower the freezing point. Impure solids commonly melt over a range and at a lower temperature than the corresponding pure solid. Not every possible impurity has the same effect.

Make a fair comparison: use the same pressure, comparable apparatus and a controlled method. Record repeated temperature readings and do not invent experimental results.

Heating investigations belong in a supervised laboratory. Learners can explore supplied data instead. Never taste laboratory samples, touch hot containers or heat sealed containers.

Think it through

A claim says 'all impurities lower boiling points'. Correct the claim and state the conditions that make your comparison fair.

4. Physical or chemical?

Classify changes using evidence of new substances.

A physical change alters state, shape, size or distribution without making a new chemical substance. Melting ice, cutting paper and dissolving sugar in water are examples.

A chemical change forms one or more new substances. Rusting iron, burning wood and cooking an egg involve chemical changes.

Reversibility is a clue, not the definition. Cutting paper is physical even though putting the original sheet back together is difficult. Some chemical reactions can also be reversed under suitable conditions.

Gas production, colour change, a precipitate, light or temperature change can suggest a reaction, but a single observation is not always proof. Boiling produces bubbles without forming a new substance.

Compare what substances existed before and after the change. Use several observations and appropriate scientific knowledge; do not identify unknown substances by tasting or directly smelling them.

Think it through

A friend says cutting paper must be chemical because it is hard to reverse. Give a better test and explain your classification.

5. Science in daily life

Apply state-change ideas safely and explain useful changes.

Wet clothes dry as water evaporates. Air movement, a larger exposed surface and suitable warmer conditions can increase drying rate.

Sweat evaporation transfers energy away from the skin and helps cool the body. Drink safe water and avoid unsafe heat exposure; this is an explanation, not a medical treatment.

Freezing food slows many processes involved in spoilage, but does not sterilise food. Safe storage and handling remain essential.

Metal casting uses melting followed by solidification. This is industrial work, not a home experiment. Cooking can involve both physical state changes and chemical changes.

Try a safe observation: put an ice cube in a clean cup on a tray at room temperature. Record its appearance every two minutes. Do not use flames, boiling water or sealed containers. Explain what changes and what stays the same.

Think it through

Design a fair comparison of drying on two cloths. Change one factor, keep the others similar, record observations and explain the particles' role.

Revision cards

Solid

Particles are close together and vibrate around fixed positions.

Liquid

Particles are close together but can move past one another.

Gas

Particles are far apart and move randomly through the container.

Diffusion

Net spreading from higher to lower concentration due to random motion.

Melting / freezing

Solid to liquid / liquid to solid.

Evaporation / condensation

Liquid to gas at the surface / gas to liquid.

Heating-curve plateau

A pure substance changes state at steady temperature under fixed pressure.

Physical change

No new chemical substance forms. Reversibility alone is not the test.

Chemical change

One or more new substances form.

Impurity caveat

Dissolved non-volatile solutes generally raise boiling point at the same pressure.

100-question practice bank

40 multiple-choice questions and 60 short-answer questions. This is a practice bank, not a standardised exam. Give reasons wherever possible. Answer key follows the questions.

Particles & states

1. Which state normally has a fixed shape?

A. Liquid

B. Gas

C. All three

D. Solid

2. Which state fills all the available space in a container?

A. Liquid

B. Ice only

C. Gas

D. Solid

3. What do particles in a solid do?

A. Disappear

B. Vibrate around fixed positions

C. Remain completely still

D. Travel freely across the room

4. Why is a gas relatively easy to compress?

A. There is much space between its particles

B. Its particles are soft

C. Its particles shrink

D. It has no particles

5. Which description best fits a liquid?

A. Fixed shape but no fixed volume

B. No mass

C. Widely separated particles

D. Fixed volume but no fixed shape

6. Ink spreads through still water mainly because of what process?

A. Condensation

B. Rusting

C. Diffusion

D. Freezing

7. When temperature rises, average particle kinetic energy generally does what?

A. Turns into mass

B. Increases

C. Decreases

D. Becomes zero

8. During thermal expansion, what mainly changes?

A. Average particle separation

B. Size of each particle

C. Number of atoms in each particle

D. Identity of the substance

9. Give the three common states of matter.

10. Why does a liquid take the shape of its container?

11. Why does a solid retain its shape?

12. Describe gas particle motion.

13. Does diffusion require stirring? Explain.

14. State the overall concentration direction of diffusion.

15. Why is water difficult to compress?

16. Does a solid's particle motion stop at room temperature?

17. Predict what happens to diffusion when temperature increases, all else equal.

18. Why can a smell spread across a room without a fan?

19. Name one particle-model limitation.

20. Explain why pouring water changes its shape but not its identity.

Changes of state

21. What is the change from solid to liquid called?

A. Freezing

B. Condensation

C. Boiling

D. Melting

22. What is the change from gas to liquid called?

A. Freezing

B. Evaporation

C. Condensation

D. Melting

23. Where does evaporation occur?

A. Only at 100 degrees Celsius

B. At the surface of a liquid

C. Only at the bottom

D. Only inside a solid

24. What does a flat melting section on an ideal heating curve show?

A. A state change at constant temperature

B. No energy entering

C. Particles disappearing

D. A chemical reaction in all cases

25. At standard atmospheric pressure, pure water boils at approximately what temperature?

A. 0 degrees Celsius

B. 25 degrees Celsius

C. Always 80 degrees Celsius

D. 100 degrees Celsius

26. Which statement distinguishes boiling from evaporation?

A. Boiling happens only at the surface

B. Evaporation is always chemical

C. Boiling occurs throughout the liquid

D. Evaporation requires bubbles throughout

27. Which state change releases energy to the surroundings?

A. Boiling

B. Freezing

C. Melting

D. Evaporation

28. Why can water boil below 100 degrees Celsius on a high mountain?

A. Atmospheric pressure is lower

B. Water stops being water

C. The particles are larger

D. Gravity supplies heat

29. Name the change from liquid to solid.

30. Name the change from liquid to gas below boiling point.

31. Why is melting ice a physical change?

32. What two states coexist during melting?

33. What two states coexist during boiling?

34. What does a rising section of a heating curve indicate?

35. Why does temperature stay steady during ideal melting?

36. What should label the vertical axis of a heating curve?

37. Name a suitable horizontal-axis quantity for a heating curve.

38. Explain water droplets on the outside of a cold cup.

39. Does evaporation require the whole liquid to reach boiling point?

40. Why must pressure be stated when comparing boiling points?

Purity & temperature

41. What does a non-volatile dissolved solute generally do to water's boiling point?

A. Always lowers it

B. Has no possible effect

C. Makes it exactly zero

D. Raises it at the same pressure

42. Salt dissolved in water generally does what to its freezing point?

A. Makes water flammable

B. Changes water into salt

C. Lowers it

D. Always raises it

43. An impure solid often melts in what way?

A. Without any energy transfer

B. Over a range of temperatures

C. At every temperature equally

D. Only at 100 degrees Celsius

44. Which condition must match in a fair boiling-point comparison?

A. External pressure

B. Label colour

C. Student's name

D. Day of the week only

45. Which is a suitable safety rule?

A. Heat a sealed bottle

B. Taste the sample

C. Hold hot glass with bare hands

D. Use teacher-supervised heating

46. A pure substance is best described as what?

A. Any drinking water

B. Any white powder

C. A single chemical substance

D. Anything that looks clear

47. Why should measurements be repeated?

A. To avoid recording data

B. To check consistency and reduce the influence of random variation

C. To guarantee a preferred answer

D. To change the substance's identity

48. Which impurity condition supports the usual boiling-point elevation claim?

A. The solute is dissolved and non-volatile

B. Any solid lying nearby

C. Any gas above the container

D. All impurities without exception

49. Define a mixture.

50. Why is clear water not necessarily pure?

51. Give one non-volatile solute commonly used in classroom examples.

52. Correct: 'Every impurity lowers the boiling point.'

53. Give a reason not to compare boiling water in two places at different altitudes.

54. Why should a thermometer be read consistently?

55. Why should you never taste laboratory water samples?

56. Why must a heated container not be sealed?

57. Which is generally lower: the freezing point of pure water or salt water at the same pressure?

58. Why is 'all impurities behave identically' a weak claim?

59. What should a learner do if measured data disagree with a prediction?

60. How can learners study boiling-point effects without heating anything?

Physical or chemical?

61. What is the defining feature of a chemical change?

A. It is always fast

B. It is always reversible

C. Its shape changes

D. New substances form

62. Cutting paper is usually classified as what?

A. Condensation

B. Combustion

C. A physical change

D. A chemical change

63. Rusting iron is what type of change?

A. Freezing

B. Chemical

C. Physical only

D. Melting

64. Bubbles in boiling water prove what?

A. Water vapour is forming, not necessarily a reaction

B. A new element has formed

C. Water is burning

D. All bubbles mean chemical change

65. Which is a chemical change?

A. Breaking glass

B. Melting wax without burning

C. Crushing chalk

D. Cooking an egg

66. Which is the best classification test?

A. Ask only whether it is loud

B. Ask whether a tool was used

C. Ask whether new substances formed

D. Ask only whether it can be reversed

67. Dissolving sugar in water is usually what?

A. Combustion

B. Physical

C. Chemical in every case

D. Nuclear

68. Which observation alone is insufficient to prove a chemical change?

A. Colour change

B. Confirmed new substance formation

C. Confirmed change in chemical composition

D. Identification of reaction products

69. Define a physical change.

70. Give a chemical-change example involving iron.

71. Why is crushing chalk physical?

72. Why is burning wood chemical?

73. Why is reversibility alone an unreliable test?

74. Give a physical change that is difficult to reverse.

75. Name two possible signs of a chemical reaction.

76. Explain why melting wax and burning wax are different.

77. Is a temperature change always evidence of a chemical reaction?

78. Why must you not smell an unknown reaction mixture directly?

79. What evidence strengthens a claim of chemical change?

80. Does a change in shape necessarily mean a chemical change?

Science in daily life

81. Which process dries wet clothes?

A. Freezing

B. Rusting

C. Condensation

D. Evaporation

82. How does evaporating sweat help the body?

A. It stops all heat transfer

B. It turns salt into water

C. It transfers energy away from the skin

D. It creates new skin

83. What does freezing food do?

A. Removes every impurity

B. Slows many spoilage processes without sterilising it

C. Kills every microorganism

D. Makes all food safe forever

84. Which sequence is used in metal casting?

A. Melting then solidification

B. Rusting then evaporation

C. Condensation then burning

D. Diffusion then digestion

85. Which activity is suitable for the unsupervised observation in this lesson?

A. Boiling salt water

B. Burning plastic

C. Heating a sealed bottle

D. Watching ice melt at room temperature

86. What usually helps a wet cloth dry faster, all else equal?

A. Sealing it in a wet bag

B. Reducing airflow

C. A larger exposed surface area

D. Folding it into a tight ball

87. What makes an experiment a fair comparison?

A. Never recording a method

B. Changing one tested factor while controlling others

C. Changing everything

D. Choosing only preferred readings

88. Which is a chemical change used in everyday food preparation?

A. Baking a cake

B. Melting ice

C. Freezing water

D. Cutting a cucumber only

89. Why can a breeze help clothes dry?

90. Why can spreading a wet cloth help it dry?

91. Why is a freezer not a steriliser?

92. Identify a useful application of solidification.

93. What should be recorded during the ice-cube observation?

94. What remains chemically the same when the ice melts?

95. Suggest one variable to test in a cloth-drying investigation.

96. Name two conditions to keep similar when comparing drying surface area.

97. How could you share a learning without sharing private data?

98. What is the purpose of comparing observations with a partner?

99. Give an example of physical and chemical changes in the same activity.

100. Write a useful exit-ticket question for this topic.

Answer key & explanations

Short-answer responses are examples, not the only acceptable wording. Award credit for scientifically correct explanations. In the HTML, these are self-assessed, not AI-marked.

1. D. Solid

Solid particles vibrate around fixed positions.

2. C. Gas

Gas particles move randomly through the available space.

3. B. Vibrate around fixed positions

A solid's particles move, but remain near fixed positions.

4. A. There is much space between its particles

Compression mainly reduces the gaps between gas particles.

5. D. Fixed volume but no fixed shape

A liquid flows while retaining approximately the same volume.

6. C. Diffusion

Random particle motion produces net spreading.

7. B. Increases

Temperature is related to average particle kinetic energy.

8. A. Average particle separation

Expansion is not caused by individual particles becoming larger.

9. Solid, liquid and gas.

Water can occur as ice, liquid water and water vapour.

10. Its particles can move past one another.

The particles stay close but are not locked into fixed positions.

11. Its particles remain near fixed positions, held by strong attractions.

Vibration does not normally let the whole structure flow.

12. Particles move rapidly and randomly, colliding with one another and container walls.

They are not arranged in a fixed pattern.

13. No. Particles move randomly even without stirring.

Stirring can mix substances faster, but is not the definition of diffusion.

14. From higher concentration towards lower concentration.

Individual particles move in many directions; this is the net effect.

15. Its particles are already close together with little empty space between them.

The particles are not arranged far apart as in a gas.

16. No. Its particles continue to vibrate.

A fixed shape does not mean motionless particles.

17. It generally becomes faster.

Particles have greater average kinetic energy.

18. Vapour particles spread by diffusion; air currents can also help.

Do not deliberately smell unknown laboratory substances.

19. Diagrams are not to scale and do not show every particle or every interaction.

A model explains selected features rather than being a literal photograph.

20. Particles rearrange and flow; the substance remains water.

A shape change alone does not create a new substance.

21. D. Melting

Melting occurs when a solid becomes a liquid.

22. C. Condensation

Gas particles come closer together as liquid forms.

23. B. At the surface of a liquid

Evaporation can occur below the boiling point.

24. A. A state change at constant temperature

Energy changes the arrangement of particles during melting.

25. D. 100 degrees Celsius

The boiling point depends on external pressure.

26. C. Boiling occurs throughout the liquid

Boiling involves vapour bubbles forming and growing within the liquid.

27. B. Freezing

The substance releases energy as liquid changes to solid.

28. A. Atmospheric pressure is lower

Lower external pressure lowers water's boiling point.

29. Freezing or solidification.

Particles become arranged in a solid structure.

30. Evaporation.

It happens at the liquid surface.

31. The substance is still water; no new substance is formed.

Chemical identity, not appearance alone, is the key.

32. Solid and liquid.

Both are present until the solid has melted.

33. Liquid and gas.

Energy continues to transfer while vapour forms.

34. Temperature is rising within a single state.

Average particle kinetic energy increases.

35. Supplied energy changes particle arrangement rather than raising average kinetic energy.

Energy is still being transferred.

36. Temperature, with its unit such as degrees Celsius.

Axes need both quantity and unit where appropriate.

37. Energy supplied; time can be used if the heating rate is controlled.

The demo's horizontal axis is energy supplied.

38. Water vapour in nearby air cools and condenses on the cup.

Water has not passed through the cup wall.

39. No.

Surface particles can escape at lower temperatures.

40. A substance's boiling point changes with external pressure.

Comparisons should be made under the same pressure conditions.

41. D. Raises it at the same pressure

The statement applies to suitable dissolved non-volatile solutes.

42. C. Lowers it

This is freezing-point depression.

43. B. Over a range of temperatures

It commonly melts over a broader, lower range than the pure solid.

44. A. External pressure

Pressure itself affects boiling point.

45. D. Use teacher-supervised heating

Hot equipment and liquids can cause serious burns.

46. C. A single chemical substance

Appearance alone cannot establish purity.

47. B. To check consistency and reduce the influence of random variation

Repeated trials help assess reliability.

48. A. The solute is dissolved and non-volatile

The conditions matter; avoid an unlimited generalisation.

49. A material containing two or more substances physically combined.

Its components have not necessarily reacted to form a new substance.

50. It can contain dissolved substances that are not visible.

Transparency does not prove chemical purity.

51. Table salt dissolved in water.

Use a supervised demonstration or supplied data rather than tasting.

52. A dissolved non-volatile solute generally raises the solvent's boiling point at the same pressure.

Different mixtures can behave differently.

53. Different atmospheric pressures may change boiling temperatures.

Pressure would be a confounding variable.

54. Different reading methods can introduce measurement errors.

Follow the instrument's instructions and use the same method.

55. They may contain harmful or unknown substances.

Labels and clarity do not guarantee safety.

56. Gas or vapour pressure can build up and rupture the container.

Use only teacher-approved apparatus and methods.

57. Salt water's freezing point.

Suitable dissolved salt causes freezing-point depression.

58. The effect depends on the substances, concentration and conditions.

Scientific statements need appropriate limits.

59. Record the data honestly, check the method and repeat where appropriate.

Do not alter results to match the prediction.

60. Interpret teacher-supplied data or use a suitable simulation.

Data-based activities avoid hot-liquid hazards.

61. D. New substances form

Chemical identity changes when new substances form.

62. C. A physical change

Changing size does not by itself create a new substance.

63. B. Chemical

Rust contains substances different from the original iron.

64. A. Water vapour is forming, not necessarily a reaction

Boiling is a physical state change.

65. D. Cooking an egg

Cooking changes substances in the egg.

66. C. Ask whether new substances formed

Reversibility alone is not reliable.

67. B. Physical

Sugar remains sugar and can be recovered by appropriate separation.

68. A. Colour change

Mixing coloured materials can change appearance without a reaction.

69. A change that does not create a new chemical substance.

State, shape, size or distribution can change.

70. Rusting.

Iron reacts in the presence of oxygen and water.

71. The pieces become smaller, but the substance remains chalk.

Size is not chemical identity.

72. New substances such as gases and ash form.

Combustion is not just a state change.

73. Some physical changes are hard to reverse, and some chemical reactions can be reversed.

The formation of new substances is the defining distinction.

74. Cutting paper or breaking glass.

The substance can remain chemically the same.

75. For example, gas production and formation of a precipitate.

These clues require context and do not individually prove every reaction.

76. Melting is a physical state change; burning forms new substances.

A burning candle can show both kinds of change.

77. No. Physical processes can also transfer energy and change temperature.

Consider the substances before and after.

78. It may release harmful vapours.

Follow teacher-approved safety procedures.

79. Reliable identification of products that were not present initially.

Multiple observations and chemical knowledge support classification.

80. No.

Bending a metal wire changes shape without necessarily forming new substances.

81. D. Evaporation

Liquid water escapes as water vapour.

82. C. It transfers energy away from the skin

Evaporation needs energy.

83. B. Slows many spoilage processes without sterilising it

Safe handling is still necessary.

84. A. Melting then solidification

Molten metal is shaped and allowed to solidify.

85. D. Watching ice melt at room temperature

The ice observation avoids flames and hot liquids.

86. C. A larger exposed surface area

More water is exposed to air.

87. B. Changing one tested factor while controlling others

Controlled conditions help interpret differences.

88. A. Baking a cake

Baking involves reactions that form new substances.

89. It removes humid air near the fabric and helps continued evaporation.

Airflow is one factor; humidity and temperature also matter.

90. A larger wet surface is exposed to air.

Compare similar cloths and initial amounts of water.

91. Some microorganisms can survive freezing and become active after thawing.

Observe safe food-handling guidance.

92. Making an ice cube or allowing cast metal to set.

The substance changes from liquid to solid.

93. Time, appearance and evidence of liquid water forming.

Use regular intervals and avoid inventing measurements.

94. The substance remains water.

Its state and particle arrangement change.

95. For example, exposed surface area.

Choose one variable rather than changing several at once.

96. For example, cloth material and starting amount of water; also keep temperature and airflow similar.

Different conditions could explain different drying times.

97. Share a topic summary or lesson file without student names or personal scores.

Learning content can be shared separately from personal information.

98. To explain reasoning, notice differences and improve evidence-based conclusions.

Respectful discussion is more useful than just copying answers.

99. A candle's wax melts physically while wax vapour burns chemically.

Discuss the example without lighting a flame.

100. For example: why is melting ice physical but rusting iron chemical?

A good question asks for reasoning about new substances.

Explore & share responsibly

The HTML has five original visual missions, each with a prediction challenge. The purity mission compares pure water with a salt solution. Its exaggerated temperature gap is conceptual, not a measurement or a prediction of real heating times. Use the salt toggle to compare with two pure-water samples.

Optional click-to-load YouTube embeds: 'Boiling Point Demo Video - Water vs Salt Water' by Matthew Robinson / Holly Grove Middle School (youtube.com/watch?v=ym1ggwU3lDY), and 'Boiling Point Elevation | Colligative Properties' by The Science Classroom (youtube.com/watch?v=cBkvORRSSuo). Titles and publishers were verified using YouTube metadata, but the full videos have not been reviewed for classroom suitability. The Short includes extension vocabulary.

Videos need internet and may include adverts or tracking. Embedded playback may fail when opening local HTML files; direct YouTube links remain available. No videos autoplay or load before the learner chooses Load. A teacher should review videos and search results before classroom use.

Suggested searches: states of matter and particle model; heating curves and changes of state; physical and chemical changes; effects of salt on boiling and freezing.

Share this original PDF as a revision handout. To share the interactive lessons offline, send the complete site folder, keeping index.html and both topic pages with their revision PDFs. To share by web link, host the folder on a static website with index.html at its root. The source curriculum stays an external link; do not include it in the upload. No accounts, analytics or server-side student records are included.

Local progress belongs to the current browser and device. It is not a class record, a verified grade, or a secure backup. The HTML includes a reset button and a progress export.

Teacher review checklist

Check scientific accuracy, curriculum fit, accessibility, question difficulty, practical safety and any linked resources before classroom use. Extend this prototype into supervised investigations and assessment across the full 22-lesson sub-strand.

Source: KICD Integrated Science Grade 8, revised 2024; source page references on the cover.